

Discrete directed graphical models

$$p(x_1 \dots x_n) \equiv p(x), \quad x_i \in \{1 \dots k_i\}$$

In the discrete case, we can think about the joint distribution as an n -dimensional probability table that sums to 1 over all cells

$$\sum_{x_1} \dots \sum_{x_n} p(x_1 \dots x_n) = 1$$

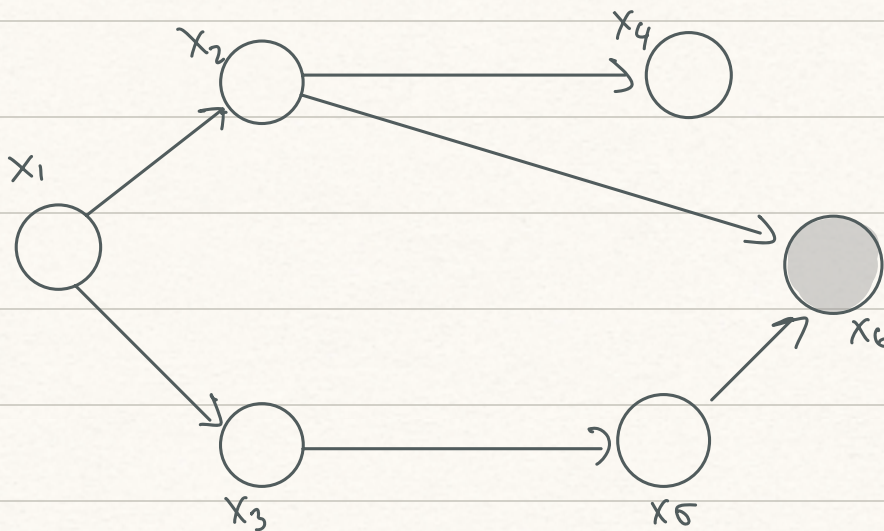
The number of cells in this table is exponential in the number of nodes

$$|p(\dots)| = \prod_i k_i, \quad \text{e.g. } k^n \text{ if } k_i = k \quad \forall i$$

A directed graphical model expresses the joint distribution as **local probability table**

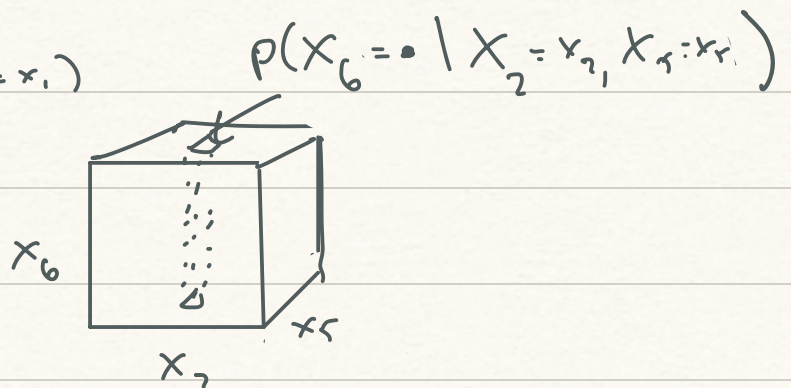
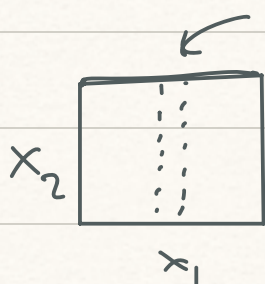
$$p(x_1 \dots x_n) = \prod_{i=1}^n p(x_i | x_{\text{par}(i)})$$

Where $\text{par}(i)$ is the set of parents of node i in the graph. Note that this factorization depends on the topological ordering of nodes $i = 1 \dots n$. Taking the conditional independences into account, we can store the joint distribution using fewer parameters via **local probability tables (LPTs)**. Let's see an example:



$$p(x_1 \dots x_6) = p(x_1) p(x_2 | x_1) \dots p(x_6 | x_2, x_5)$$

local probability tables



As we see, when there are conditional independences in the model, the total number of parameters we need to represent the joint distribution is just the total number of cells in the local probability tables:

$$\text{total \# cells : } \sum_{i=1}^n k_i \prod_{j \in \text{par}(i)} k_j$$

Say that $k_1 = \dots = k_n = K$, then compare this to how many cells there are in the full table:

$$\sum_{i=1}^n K^{|\text{par}(i)|} \quad \text{vs} \quad K^n$$

So it goes from being exponential in the number of nodes, to exponential in the size of the largest parent set. In this special case of trees, every node has only one parent, in which case:

$$nK \quad \text{vs} \quad K^n$$

So graphical models formalize how joint distributions factorize, which we can exploit to fit and query models efficiently. We will explore how to exploit these properties to do efficient exact inference in discrete graphical models today.

Evidence nodes vs. query nodes

Today we will think about performing queries efficiently. A query is just a marginal distribution, and performing a query means computing that marginal. Define the following:

① X_E the set of "evidence nodes": these are the variables whose value we will condition on

② X_F the set of "query nodes": these are the variables whose marginal distribution given the evidence nodes we seek to compute

③ $X_R = (X_E \cup X_F)^c$ all other variables in the graph that aren't evidence or query nodes

$$\text{Goal: } P(X_F | X_E) = \frac{\sum_{x_R} P(x)}{\sum_{x_F} \sum_{x_R} P(x)}$$

The goal will be to compute this marginal. This involves marginalizing over all variables that are not evidence or query nodes, which would be exponential in the number of them if done naively

e.g. $O(K^{|R|})$ naively if $k_1 = \dots = k_n = K$

We will use further **special bar notation to denote values that are conditioned on**. Let's compute:

$$\text{compute } p(x_1 | \bar{x}_6) = p(X_1 = x_1 | X_6 = \bar{x}_6)$$

Before expanding this, we will define a useful function which is the **"evidence potential"**:

$$\delta_{\bar{x}_6}(x_6) = \begin{cases} 1 & \text{if } x_6 = \bar{x}_6 \\ 0 & \text{if } x_6 \neq \bar{x}_6 \end{cases}$$

For all evidence nodes, we multiply the joint by their evidence potential. This lets us write any conditioning operation as simply summing over those variables:

$$p(x_1, \bar{x}_6) = \sum_{x_2} \cdots \sum_{x_6} p(x_1) p(x_2 | x_1) \cdots p(x_6 | x_2, x_5) \delta_{\bar{x}_6}(x_6)$$

Some of the sums push in, since not all local probabilities involve all variables:

$$= p(x_1) \sum_{x_2} p(x_2 | x_1) \underbrace{\sum_{x_3} \cdots \sum_{x_6} p(x_6 | x_2, x_5) \delta_{\bar{x}_6}(x_6)}_{\triangleq m_6(x_2, x_5)}$$

Summing out or "eliminating" x_6 first reveal an **intermediate factor**, which is a function of x_2 and x_5 . We will interpret this factor as a **"message"** from the eliminated variable x_6 . (The message interpretation will become clear later.)

$$= p(x_1) \sum_{x_2} \cdots \underbrace{\sum_{x_5} p(x_5 | x_3) m_6(x_2, x_5)}_{\triangleq m_5(x_2, x_3)}$$

Eliminating x_5 next reveals a factor involving only x_2 and x_3 . Since it does not depend on x_4 , it then pushes past that sum:

$$= p(x_1) \cdots \underbrace{\sum_{x_3} p(x_3 | x_1) m_5(x_2, x_3)}_{\triangleq m_3(x_1, x_2)} \underbrace{\sum_{x_4} p(x_4 | x_2)}_{\triangleq m_4(x_2) = 1}$$

We continue by eliminating x_4, x_3, x_2 and the final form is just $p(x_1)$ x the last message from x_2 :

$$= p(x_1) \underbrace{\sum_{x_2} p(x_2 | x_1) m_3(x_1, x_2) m_4(x_2)}_{\triangleq m_2(x_1)}$$

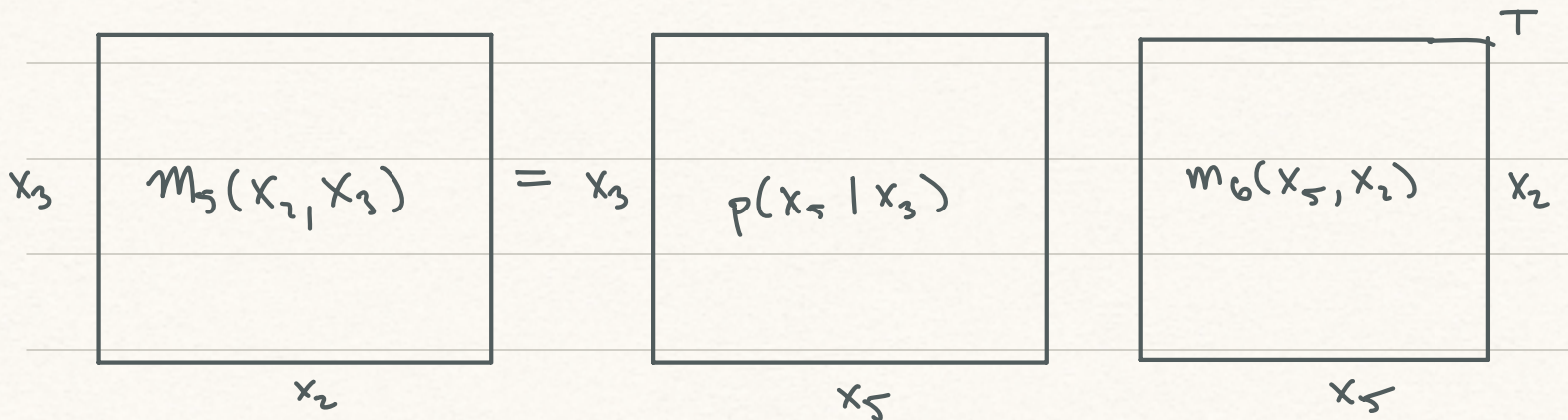
We can thus compute our query as:

$$p(x_1 | \bar{x}_6) = \frac{p(x_1) m_2(x_1)}{\sum_j p(x_1 = j) m_2(x_1 = j)}$$

Comments:

- Iterative **sum-product** procedure
- Eliminated variables left behind **messages** (intermediate factors)
- Each message is a function of non-eliminated variables
- We can think about each message as specific kind of **probability table**

$$\text{e.g. } m_5(x_2, x_3) = \sum_{x_5} p(x_5 | x_3) m_6(x_5, x_2)$$



What exactly is the message computing?

$$\begin{aligned} m_5(x_2, x_3) &= \sum_{x_5} p(x_5 | x_3) \underbrace{\sum_{x_6} p(x_6 | x_2, x_5) \delta_{\bar{x}_6}(x_6)}_{= p(\bar{x}_6 | x_2, x_5)} \\ &= \underbrace{\sum_{x_5} p(x_5 | x_3) p(\bar{x}_6 | x_2, x_5)}_{= p(\bar{x}_6 | x_2, x_3)} \\ &= p(\bar{x}_6 | x_2, x_3) \end{aligned}$$

Answer: a table of conditional evidence probabilities. These are probabilities of evidence nodes given non-eliminated nodes.

but not always!

What if we had instead eliminated x_2 first?

$$m_2(\dots) = \sum_{x_2} p(x_2 | x_1) p(x_4 | x_2) p(x_6 | x_2, x_5)$$

This would have created an intermediate factor of 4 nodes, which would take many more cells of a table to store:

$$= m_2(x_1, x_4, x_5, x_6) \quad \text{4 terms}$$

So **the order of elimination matters**. To see this, let's first formalize the variable elimination algorithm, which is a general algorithm for computing any marginal in a graphical model.

The Variable Elimination algorithm for directed graphical models

Input: graph G , target query $p(x_F | \bar{x}_E)$

① ORDER $(x_1 \dots x_n)$ s.t. x_F is last

② INITIALIZE "ACTIVE LIST" of functions as:

[^] abuse of notation, this is a final thing to pass, but is a set

$$\lambda = \left[p(x_i | x_{\text{par}(i)}) \text{ for } i = 1 \dots n \right] \\ + \left[\delta_{\bar{x}_i}(x_i) \text{ for } i \in E \right]$$

③ for i in ORDER:

- $\lambda_i = [\text{all } f(x_i, \dots) \text{ in } \lambda]$

[^] eg, all the things in the active list for which x_i is an input

- $S_i = [\text{all } x_j \text{ s.t. } f(x_j, x_i, \dots) \text{ in } \lambda_i]$

- $\phi_i(x_i, S_i) = \prod_{f \in \lambda_i} f(x_i, \dots)$
^{subset of S_i}

define the message from x_i

- $m_i(S_i) = \sum_{x_i} \phi_i(x_i, S_i)$

remove functions from the active set

- $\lambda = \lambda - \lambda_i$

add message from x_i to active set

- $\lambda = \lambda + m_i(S_i)$

below is typically the total distribution of evidence given x_F , divided by the total across x_F

④ RETURN: $\frac{\phi_F(x_F, S_F)}{m_F(S_F)} \equiv \frac{p(x_F, \bar{x}_E)}{\sum_{x_F} p(x_F, \bar{x}_E)}$

[^] abuse of notation, should not be a function of S_F anymore?

Comments:

- notice that the complexity depends on the sizes of the intermediate sets S_i that are created when eliminating each x_i
- S_i is called the "elimination clique"
- Different orderings $ORDER(x_1 \dots x_n)$ lead to different sequences of S_i 's, and thus different complexity
- It's generally intractable to determine the optimal ordering. However for trees there is a known answer.

remove functions from the active

The Variable Elimination algorithm for trees

The joint distribution for a tree-structured graphical model can always be written in terms of the probability of the **root node x_r** times a product over all edges:

$$p(x_1 \dots x_n) = p(x_r) \prod_{\substack{i \rightarrow j \\ \in G}} p(x_j | x_i)$$

For defining a general-purpose algorithm define the following "singleton potentials" for all nodes x_i as:

$$\psi(x_i) = \begin{cases} 1 & \text{if } x_i \notin X_E \\ \delta_{\bar{x}_i}(x_i) & \text{if } \cancel{x_i \in X_E} \quad x_i \in X_E \end{cases}$$

For only the **root node**, overwrite the definition to be:

$$\psi(x_r) = p(x_r) \psi(x_r)$$

times $1^{\wedge}$

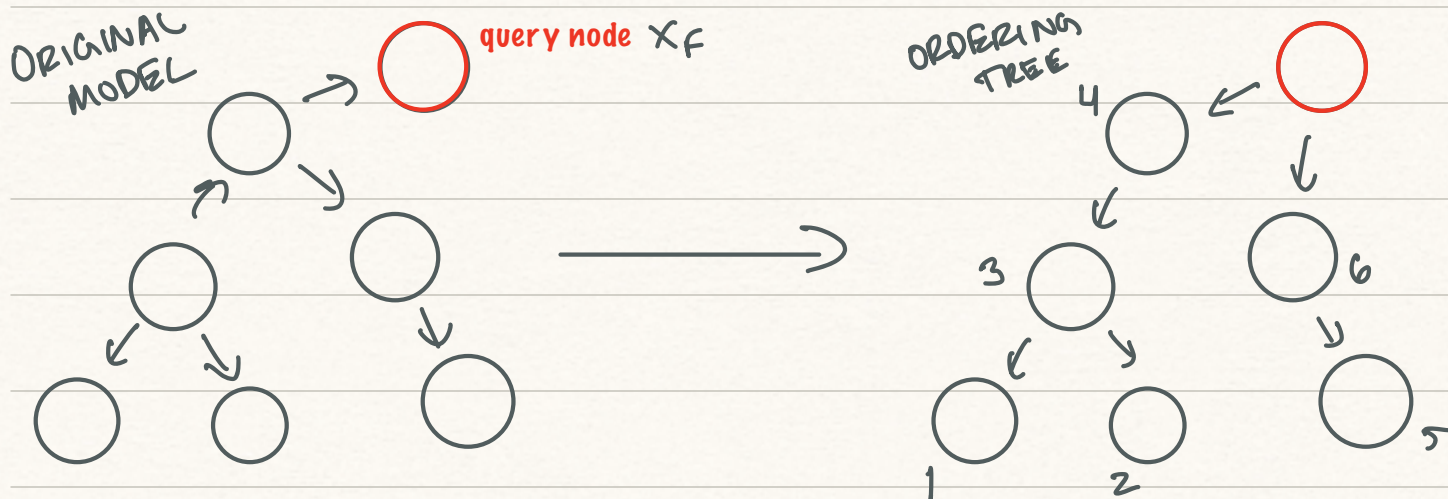
Then define the following "pairwise potentials" as:

$$\psi(x_i, x_j) = p(x_j | x_i) \quad \forall \quad i \rightarrow j \in G$$

^ parent here!

The VE algorithm for trees can then be expressed as follows.

First, $ORDER(x_1 \dots x_n)$ variables by 1) creating a new tree with the **query node at the root** and all arrows pointing away from it, and 2) **order nodes by depth-first search** in the new tree. Example:



Second, initialize the active list to be all singleton and pairwise potentials (as defined above):

$$A = [\psi(x_i, x_j) \forall i \rightarrow j] + [\psi(x_i) \forall i]$$

Third, run the for-loop as stated in the original algorithm. In this case, each message would be:

$$m_j(\dots) = \sum_{x_j} \prod_{f \in \mathcal{L}_j} f(m_j, \dots)$$

$$= \sum_{x_j} \psi(x_j) \psi(x_i, x_j) \prod_k m_k(x_j, \dots)$$

^ this distribution might not be easy to figure out, right? Like we can define trees for which the reverse is "hard" to compute

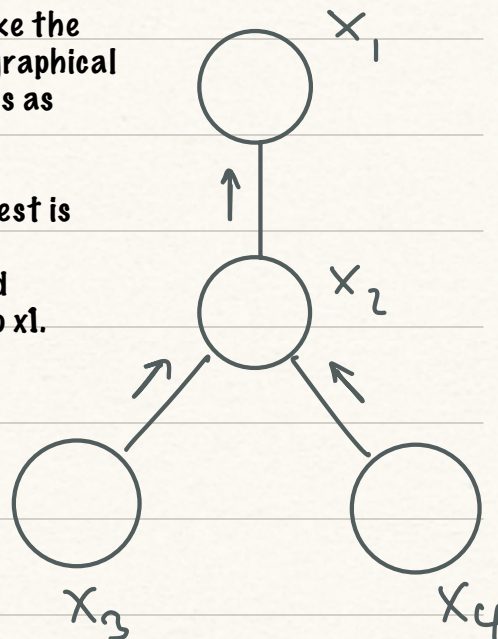
Each $m_k(\dots)$ is a message from an eliminated node that involves x_j . Due to the ordering, we are eliminating nodes bottom up (in the ordering tree, not the original model):

This means that each message only has one receiver:

$$m_k(x_j, \dots) \equiv m_{k \rightarrow j}(x_j)$$

Here's a worked example. Take the following tree-structured graphical model (which we will express as undirected).

Say that our query of interest is $P(x_1)$. In this case, variable elimination would then send messages up "up the tree" to x_1 .



$$m_{3 \rightarrow 2}(x_2) = \sum_{x_3} \psi(x_3) \psi(x_2, x_3), \quad m_{4 \rightarrow 2}(x_2) = \sum_{x_4} \dots$$

^ prob of evidence given x_2 , evidence in x_4 and its children!

$$m_{2 \rightarrow 1}(x_1) = \sum_{x_2} \psi(x_2) \psi(x_2, x_1) m_{3 \rightarrow 2}(x_2) m_{4 \rightarrow 2}(x_2)$$

^ x_1 here! ^ $\phi(x_1, x_2)$

$$p(x_1) \propto \psi(x_1) m_{2 \rightarrow 1}(x_1)$$

Now let's see the sequence of messages if instead our goal was $P(x_2)$.

Messages now flow "up the tree" to x_2 .

$$m_{1 \rightarrow 2}(x_2) = \dots$$

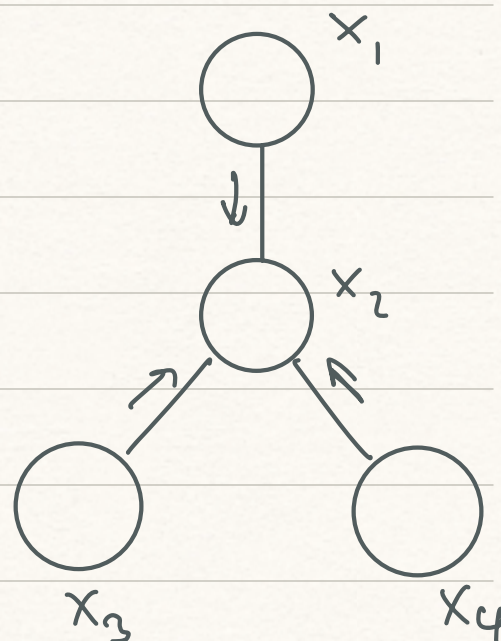
$$m_{3 \rightarrow 2}(x_2) = \dots$$

$$m_{4 \rightarrow 2}(x_2) = \dots$$

root node can be defined the same here too!

note, assuming that the evidence is held FIXED

Notice that **two of these messages are exactly the same** as the messages we generated last time when running the VE algorithm to compute $P(x_1)$. This suggests that if we want to compute the marginals for many variables, then running VE many times is wasting computation...



The sum-product algorithm

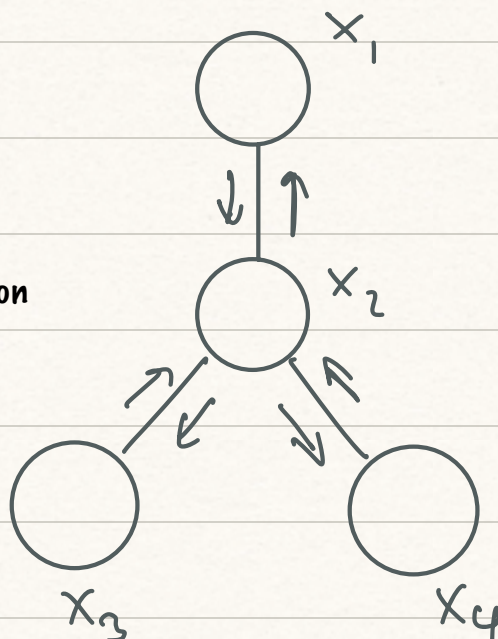
The SP algorithm is a message-passing algorithm that computes **all singleton marginals** in tree-structured graphical models using only **2 messages per edge** (i.e., the cost of 2 runs of variable elimination).

Message passing protocol: A node i sends a message to node j only after it has received messages from all other neighbors.

$$m_{i \rightarrow j}(x_j) = \sum_{x_i} \psi(x_i) \psi(x_i, x_j) \prod_{u \in \text{neigh}(x_i) \setminus x_j} m_{u \rightarrow i}(x_i)$$

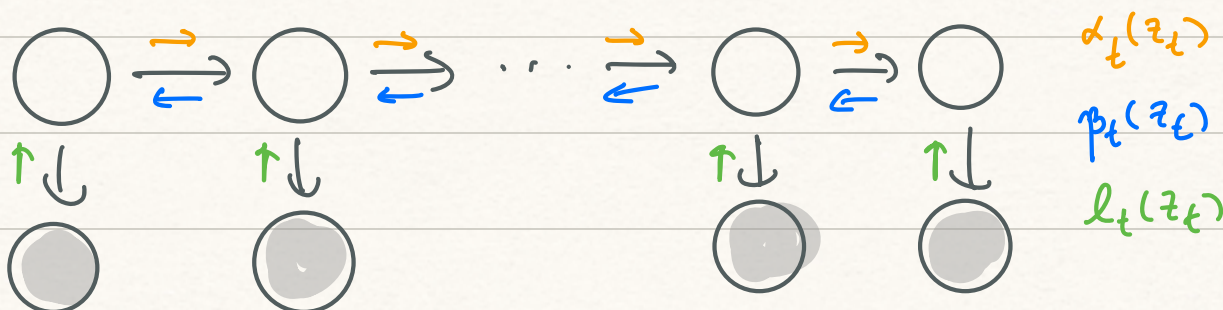
After passing all messages (two per edge), we can compute any singleton marginal as:

$$p(x_i | \bar{x}_E) \propto \psi(x_i) \prod_{j \in \text{neigh}(i)} m_{j \rightarrow i}(x_i)$$



Forwards-backwards as the sum-product algorithm

We saw last time how the forward and backward messages let us compute any singleton marginal exactly in HMMs. HMMs are trees, and forwards-backwards is just a special case of the sum-product algorithm.



Max-product algorithm

A very similar message-passing algorithm for trees is the max-product algorithm, which allows us to compute the maximum-a-posteriori (MAP) assignment of all nodes using only two sets of messages.

Goal: $X^* = \operatorname{argmax}_x p(x)$

The key idea is that the max operation distributes across the factors (just like the sum operations):

$$\begin{aligned} p(X = x^*) &= \max_x p(x) \\ &= \max_{x_1} p(x_1) \max_{x_2} p(x_2 | x_1) \dots \end{aligned}$$

Starting at the leaves, and sending messages up the tree, each node computes two messages to its parent:

$$\begin{aligned} m_{j \rightarrow i}(x_i) &= \max_{x_j} \varphi(x_j) \varphi(x_i, x_j) \prod_{k \in \operatorname{neigh}(j) \setminus x_i} m_{k \rightarrow j}(x_j) \\ m_{j \rightarrow i}^*(x_i) &= \operatorname{argmax}_{x_j} \end{aligned}$$

In other words, it sends two tables, one of which stores the maximum joint probability of descendants achievable for any input value of x_i , and the other which stores the corresponding argmax of x_j for any input value of x_i . Once the messages reach the root node, we then find the MAP by going back down the tree, starting with finding the max joint probability over all nodes, and the argmax over the root:

$$\begin{aligned} \max p(x) &= \max_{x_r} \varphi(x_r) \prod_{k \in \operatorname{neigh}(r)} m_{k \rightarrow r}(x_r) \\ X_r^* &= \operatorname{argmax}_{x_r} \end{aligned}$$

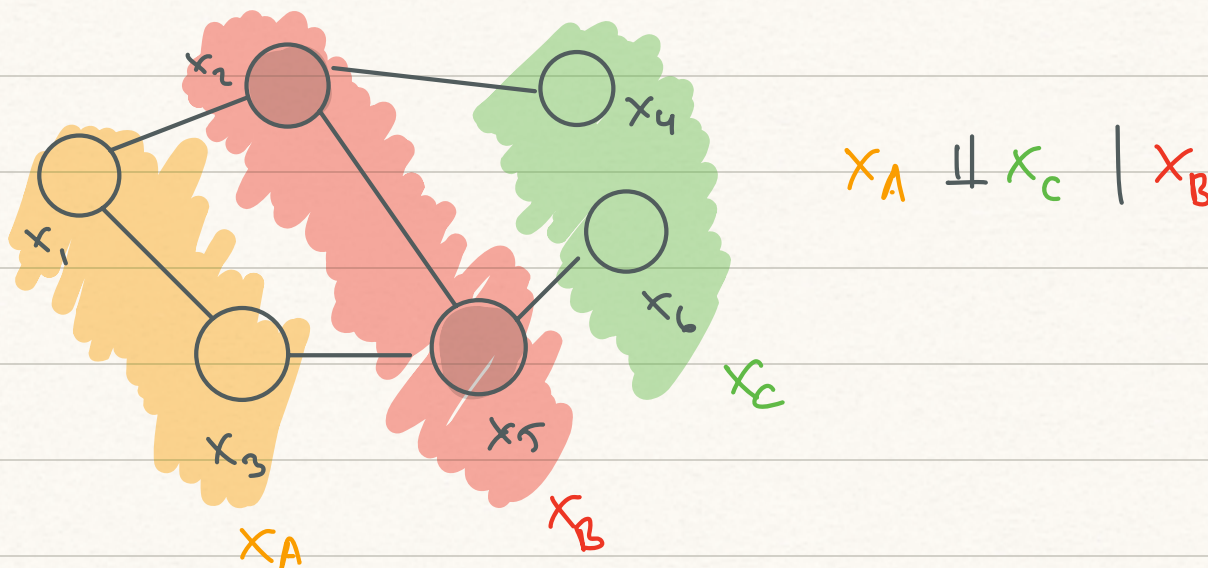
We can then assign every node, going down the tree as:

$$x_j^* = \operatorname{argmax}_{j \rightarrow i} m^*(x_i^*)$$

The well-known Viterbi algorithm is just the max-product algorithm for HMMs.

Undirected graphical models

A formalism that has been lurking in this lecture are **undirected graphical models (UGMs)**. A UGM also describes a joint distribution, but more simply: two variables are conditionally independent if there is no path between them that doesn't go through an observed variable:



UGMs are parameterized using **potential functions**:

$$p(x_1 \dots x_n) = \frac{\underbrace{k(x_1 \dots x_n)}_{\text{"kernel"}}}{\underbrace{z}_{\text{"normalizing constant" or "partition function"}}}$$

$$\Phi(x_1 \dots x_n) = \prod_{c \in C} \underbrace{\psi_c(x_c)}_{\text{potential function for "maximal clique" } c}$$

Here the product is over all **maximal cliques** in the graph (i.e., sets of nodes that are fully connected in the graph, and which would not continue to be fully connected if any other node was added).

Each potential function takes in a maximal clique and outputs a non-negative potential:

$$\psi_c(x_c) \geq 0$$

Notice this is **not a local probability**. It can be greater than 1. In order to normalize the distribution, we need to compute the **"partition function"**:

$$z = \sum_{x_1} \dots \sum_{x_n} \prod_c \psi_c(x_c)$$

Algorithms like variable elimination or sum-product can be modified to compute this efficiently in certain types of graphs. In fact, many of these algorithms were originally developed for UGMs.

UGMs are useful for modeling **spatial correlation** where there is no clear ordering between the variables.

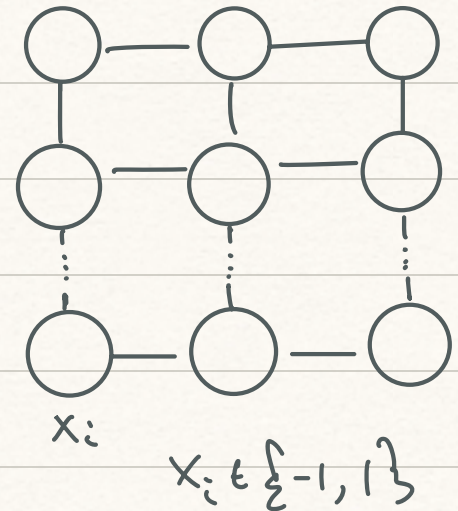
A classic example is the **Ising model** from physics, where models the "spin" of atoms arranged on a lattice.

The overall energy $H(x_1 \dots x_n)$ of a certain configuration is given by:

$$H(x_1 \dots x_n) = \sum_{i,j} x_i x_j J_{ij} \\ \equiv \varphi(x_i, x_j)$$

where J_{ij} simply says whether there is an edge between (i,j)

$$J_{ij} = \begin{cases} c & \text{if } (i,j) \in G \\ 0 & \text{otherwise} \end{cases}$$



High energy states are improbable. So the probability of any given configuration of spins is then:

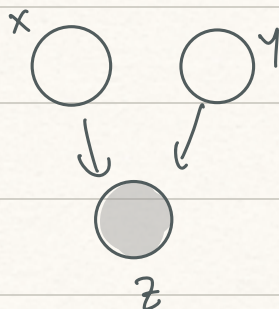
$$p(x_1 \dots x_n) \propto \exp(-H(x_1 \dots x_n))$$

A wide class of models called **energy-based models** extends this idea to arbitrary graphs.

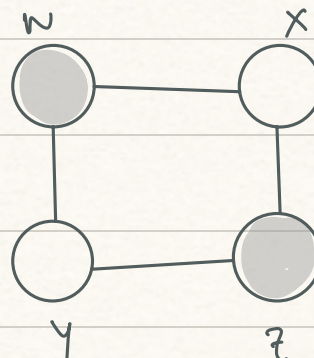
Directed vs undirected graphical models

DGMs and UGMs provide different formalisms with different strengths. With DGMs, we typically first draw the graph, and then read off the conditional independences implied by the graph (which isn't always obvious). This is useful for when we have intuitions about the generative process of our data. With UGMs on the other hand, the graph is directly defined by dependencies. This is better for when we are only aware of complex dependence (such as spatial structure) in our data.

Not all families of models (i.e., sets of conditional independences) can be expressed by both DGMs and UGMs. On the left and right are examples of sets of conditional independences that can only be expressed by DGMs and only by UGMs.



$$x \perp\!\!\!\perp y \text{ and } x \perp\!\!\!\perp y \mid z$$



$$y \perp\!\!\!\perp x \mid w, z$$

$$w \perp\!\!\!\perp z \mid x, y$$